

Infosys SP And DSE Coding Questions Analysis:

Question 1 –Easy level:

- Simple Question that can be solved by basic applications of Data Structures and Algorithms.

Question 2 – Medium Level:

Usually a question based on Greedy Algorithm

- Greedy algorithm is any algorithm that follows the problem solving heuristic of making the locally optimal choice at each stage.
- Types of greedy problems are:
 - Pure Greedy Algorithms
 - Orthogonal Greedy Algorithms
 - Relaxed Greedy Algorithms

Question 3 – Hard level:

Usually a question based on Dynamic Programming

- DP is an algorithmic technique for solving an optimisation problem by breaking it down into simpler subproblems and utilising the fact that the optimal solution to the overall problem depends upon the optimal solution to its subproblems.
- Principles of Dynamic Programming:
 - Breaking the problem down into subproblems and calculating their values. Next time, upon encountering the same subproblem, the value can be reused instead of recalculation
 - Avoid repeated work by remembering partial results, a common approach for performance enhancements.

Question 1

Problem Statement: While playing an RPG game, you were assigned to complete one of the hardest quests in this game. There are n monsters you'll need to defeat in this quest.

- Each monster i is described with two integer numbers – $power_i$ and $bonus_i$. To defeat this monster, you'll need at least $power_i$ experience points. If you try fighting this monster without having enough experience points, you lose immediately. You will also gain $bonus_i$ experience points if you defeat this monster. You can defeat monsters in any order.

- The quest turned out to be very hard – you try to defeat the monsters but keep losing repeatedly. Your friend told you that this quest is impossible to complete.

Knowing that, you're interested, what is the maximum possible number of monsters you can defeat?

(Question difficulty level: Hardest)

Input:

- The first line contains an integer, n , denoting the number of monsters. The next line contains an integer, e , denoting your initial experience.
- Each line i of the n subsequent lines (where $0 \leq i < n$) contains an integer, power_i , which represents power of the corresponding monster.
- Each line i of the n subsequent lines (where $0 \leq i < n$) contains an integer, bonus_i , which represents bonus for defeating the corresponding monster.

Input	Output	Output Description
2 123 78 130 10 0	2	- Initial experience level is 123 points. - Defeat the first monster having power of 78 and bonus of 10. Experience level is now $123+10=133$. - Defeat the second monster.
3 100 101 100 304 100 1 524	2	- Initial experience level is 100 points. - Defeat the second monster having power of 100 and bonus of 1. Experience level is now $100+1=101$. - Defeat the first monster having power of 101 and bonus of 100. Experience level is now $101+100=201$. - The third monster can't be defeated.

Sample

Input: $e=123$, $\text{monsters}=[(\text{power}=78, \text{bonus}=10), (\text{power}=130, \text{bonus}=0)]$

Output: 2

Input: $e=100$, $\text{monsters}=[(101, 100), (100, 1), (524, 304)]$

Output: 2

Question 2

Problem Statement:

Your birthday is coming soon and one of your friends, Alex, is thinking about a gift for you. He knows that you really like integer arrays with interesting properties.

He selected two numbers, N and K and decided to write down on paper all integer arrays of length K (in form $a[1], a[2], \dots, a[K]$), where every number $a[i]$ is in range from 1 to N , and, moreover, $a[i+1]$ is divisible by $a[i]$ (where $1 < i \leq K$), and give you this paper as a birthday present.

Alex is very patient, so he managed to do this. Now you're wondering, how many different arrays are written down on this paper?

Since the answer can be really large, print it modulo 10000.

Input:

- The first line contains an integer, n , denoting the maximum possible value in the arrays.
- The next line contains an integer, k , denoting the length of the arrays.

Input	Output	Output Description
2 1	2	The required length is 1, so there are only two possible arrays: [1] and [2].
2 2	3	All possible arrays are [1, 1], [1, 2], [2, 2]. [2, 1] is invalid because 1 is not divisible by 2.
3 2	5	All possible arrays are [1, 1], [1, 2], [1, 3], [2, 2], [3, 3].

Sample

$N=2, K=1 \rightarrow 2$

$N=2, K=2 \rightarrow 3$

$N=3, K=2 \rightarrow 5$

Question 3

Problem Statement:

You have an array A of N integers $A_1 A_2 \dots A_n$. Find the longest increasing subsequence $A_{i_1} A_{i_2} \dots A_{i_k}$

($1 \leq k \leq N$) that satisfies the following condition:

For every adjacent pair of numbers of the chosen subsequence $A_i[x]$ and $A_i[x+1]$ ($1 < x < k$), the expression $(A_i[x] \& A_i[x+1]) * 2 < (A_i[x] | A_i[x+1])$ is true

Note: '&' is the bitwise AND operation, '|' is the bit-wise OR operation

Input:

1. The first line contains an integer, N, denoting the number of elements in A.
2. Each line i of the N subsequent lines (where $0 \leq i < N$) contains an integer describing A_i .

Sample cases:

Input	Output	Output Description
5 15 6 5 12 1	2	One possible subsequence is: 5 12
6 9 17 2 15 5 2	2	One possible subsequence is: 2 15
7 17 16 12 2 8 17 17	3	One possible subsequence is: 2 8 17

Sample

[15,6,5,12,1] $\rightarrow$ 2 (e.g. 5, 12)
[9,17,2,15,5,2] $\rightarrow$ 2 (e.g. 2, 15)
[17,16,12,2,8,17,17] $\rightarrow$ 3 (e.g. 2, 8, 17)

Question 5

Problem Statement :

Wael is well-known for how much he loves the bitwise XOR operation, while kaito is well known for how much he loves to sum numbers, so their friend Resli decided to make up a problem that would enjoy both of them. Resil wrote down an array A of length N, an integer K and he defined a new function called Xor- sum as follows

- $\text{Xor-sum}(x) = (x \text{ XOR } A[1]) + (x \text{ XOR } A[2]) + (x \text{ XOR } A[3]) + \dots + (x \text{ XOR } A[N])$

Can you find the integer x in the range $[0, K]$ with the maximum Xor-sum (x) value?

Print only the value.

Input format

- The first line contains integer N denoting the number of elements in A.
- The next line contains an integer, k, denoting the maximum value of x.
- Each line i of the N subsequent lines (where $0 \leq i \leq N$) contains an integer describing A_i .

Constraints

- $1 \leq N \leq 10^5$
- $0 \leq K \leq 10^9$
- $0 \leq A[i] \leq 10^9$

Sample Input 1

```
1
0
989898
```

Sample Output 1

```
989898
```

Explanation:

$\text{Xor_sum}(0) = (0 \text{ XOR } 989898) = 989898$

Sample Input 2

3
7
1
6
3

Sample Output 2

14

Explanation

$\text{Xor_sum}(4) = (4^1) + (4^6) + (4^3) = 14.$

Sample Input 3

4
9
7
4
0
3

Sample Output 3

46

Explanation:

$\text{Xor_sum}(8) = (8^7) + (8^4) + (8^0) + (8^3) = 46.$

Sample

$A = [989898], K=0$	$\rightarrow 989898$
$A = [1, 6, 3], K=7$	$\rightarrow 14$
$A = [7, 4, 0, 3], K=9$	$\rightarrow 46$

Question 5

Asked in Infosys Placment Paper – Sept 2021

Problem Statement :

One of the first lessons IT students learn is the representation of natural numbers in the binary number system (base 2) This system uses only two digits, 0 and 1. In everyday life we use for convenience the decimal system (base 10) which uses ten digits, from 0 to 9. In general, we could use any numbering system.

Computer scientists often use systems based on 8 or 16. The numbering system based on K uses K digits with a value from 0 to K-1. Suppose a natural number M is given, written in the decimal system To convert it to the corresponding writing in the system based on K, we successively divide M by K until we reach a quotient that is less than K

The representation of M in the system based on K is formed by the final quotient (as first digit) and is followed by the remainder of the previous divisions For example :

- If $M=122$ and $K=8$, 122 in base 10 = 172 in base 8 This means that the number
- In decimal system = 172 in octal system.
- 172 in base 8 = $1 \cdot 8^2 + 7 \cdot 8 + 2 = 122$

You made the following observation in applying the above rule of converting natural numbers to another numbering system

- In some cases in the new representation all the digits of the number are the same. For example 63 in base 10 = 333 in base 4

Given a number M in its decimal representation, your task is find the minimum base B such that in the representation of M at base B all digits are the same.

Input Format

- The first line contains an integer, M, denoting the number given

Constraints

- $1 \leq M \leq 10^{12}$

Sample Input 1 :

41

Sample Output 1 :

40

Explanation :

Here 41 in base 40. will be 11 so it has all digits the same, and there is no smaller base satisfying the requirements

Sample Input 2 :

34430

Sample Output 2 :

312

Explanation :

Here 34430 in base 312 will have all digits the same and there is no smaller base satisfying the requirements

Question 6

Problem Statement :

You have an interesting string S of length N . It is interesting because you can rearrange the characters of this string in any order. You want to cut this string into some contiguous pieces such that after cutting, all the pieces are equal to one another.

You can't rearrange the characters in the cut pieces or join the pieces together. You want to make the number of pieces as large as possible. What is the maximum number of pieces you can get?

Note: You can observe that you may not want to cut the string at all, therefore the number of pieces is 1. Hence, the answer always exists.

Input Format

$S :: \text{STRING}$

The first line contains a string, S , denoting the string.

$\text{length}(S) :: 1 \rightarrow 2 * 10^5$

Sample Input 1:

zzzzz

Sample Output 1 :

5

Sample input 2:

ababcc

Sample Output 2:

2

Sample input 2:

abccdcabacda

Sample Output 2:

2

Question 7

Problem Statement :

Today you decided to go to the gym. You currently have energy equal to E units. There are N exercises in the gym. Each of these exercises drains A_i amount of energy from your body.

You feel tired if your energy reaches 0 or below. Calculate the minimum number of exercises you have to perform such that you become tired. Every unique exercise can only be performed at most 2 times as others also have to use the machines.

If performing all the exercises does not make you feel tired, return -1.

Input Format

$E :: \text{INTEGER}$

The first line contains an integer, E , denoting the Energy.

$E :: 1 \rightarrow 10^5$

$N :: \text{INTEGER}$

The next line contains an integer, N , denoting the number of exercises. $N :: 1 \rightarrow 10^5$

$A :: \text{INTEGER ARRAY}$

Each line i of the N subsequent lines (where $0 \leq i < N$) contains an integer describing the amount of energy drained by i -th exercise.

$A[i] :: 1 \rightarrow 10^5$

Sample Input 1:

```
6
2
1
2
```

Sample Output 1 :

```
4
```

Sample input 2:

```
10
2
```

1
2

Sample Output 2:
-1

Sample input 3:
2
3
1
5
2

Sample Output 3:
1

Sample

$E=6, A=[1,2] \rightarrow 4$
 $E=10, A=[1,2] \rightarrow -1$
 $E=2, A=[1,5,2] \rightarrow 1$

Question 8

Problem Statement :

There is a battle between heroes and villains going on. You have M heroes, all of them have the same health H . There are N villains, health of the i -th villain is V_i .

When a hero, with health H battles a villain with health V_i , one of the three scenarios can happen:

if $H > V_i$: The villain is defeated, and the health of the hero is decreased by V_i
if $H < V_i$: The villain wins, his health is not affected, and the hero is no longer able to fight. if $H = V_i$: Both are considered defeated, and neither can fight.

The heroes start fighting villains one by one in the same order, first villain 1 then villain 2 and so on. It might be possible that before defeating all the villains, all the heroes are defeated. Therefore, to ensure the victory of the heroes, you want to remove some villains from the front.

Your task is to find the minimum number of villains you need to remove from the front such that the victory of the heroes is guaranteed.

Note: If in the last battle, both the hero and villain are defeated and no more heroes or villains remain, it would still be considered a victory since all the villains are defeated.

Input Format

$N :: \text{INTEGER}$

The first line contains an integer, N , denoting the number of villains $N :: 1 \rightarrow 2 \cdot 10^5$

$M :: \text{INTEGER}$

The next line contains an integer, M , denoting the number of heroes $M :: 1 \rightarrow 2 \cdot 10^5$

$H :: \text{INTEGER}$

The next line contains an integer, H , denoting the health of each of the heroes $H :: 1 \rightarrow 10^9$

$\text{array} :: \text{INTEGER ARRAY}$

Each line i of the N subsequent lines (where $0 \leq i < N$) contains an integer

describing the health of each of the villains.

array[i] :: 1 -> 10⁹

Sample Input 1:

4
4
3
3
1
3
3

Sample Output 1 :

0

Sample input 2:

5
3
3
1
2
3
1
1

Sample Output 2:

0

Sample input 3:

5
1
4
1
2
3
1
3

Sample Output 3:

3

Sample

$E=6, A=[1,2] \rightarrow 4$
 $E=10, A=[1,2] \rightarrow -1$
 $E=2, A=[1,5,2] \rightarrow 1$